IRON WITHIN RESEARCH

RT-3 — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	21mg, 30mg	10/9/2026 · 30mg · Lot IWR-1010-RT33
4–5	21mg, 60mg	9/13/2026 · 60mg · Lot IWR-090926RT2
6–7	20mg, 32mg	9/13/2026 · 20mg · Lot IWR-090926RT1
8–9	30mg, 82mg	8/28/2026 · 30mg · Lot IWR-082526RT30
10–11	20mg, 22mg	8/28/2026 · 20mg · Lot IWR-082526RT20
12–26	6mg, 10mg, 12mg, 15mg, 22mg, 23mg, 30mg, 31mg, 76mg, 93mg	8/22/2026
27–29	10mg, 84mg	6/24/2026 · Lot IWR-5859103
30–32	9mg, 30mg	6/24/2026 · Lot IWR-4862753
33–35	5mg, 15mg	6/24/2026 · Lot IWR-1764133
36–42	5mg, 10mg, 11mg, 14mg, 16mg, 20mg, 30mg, 82mg	4/16/2026

Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0303

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 30 mg

PASS

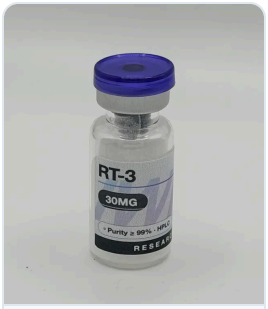
BATCH #	PURITY	MASS	IDENTITY
IWR-1010-RT33	99.91%	32.21 mg	Confirmed

COA #:	SBA-26-0303	Method:	Full QC Panel
Lot Number:	IWR-1010-RT33	Analysis Date:	2026-10-09
Labeled Content:	30 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.91%
Content (per vial)	Report only	32.21 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0303**
Access Code:	**GV2N7NXU**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0191

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 60 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926RT2	99.97%	66.21 mg	Confirmed

COA #:	SBA-26-0191	Method:	Full QC Panel
Lot Number:	IWR-090926RT2	Analysis Date:	2026-09-13
Labeled Content:	60 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Red #B8232F

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.97%
Content (per vial)	Report only	66.21 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.01 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0191**
Access Code:	**FZAUDQU3**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

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BIOANALYTICS

Certificate of Analysis

#SBA-26-0190

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TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926RT1	99.96%	21.32 mg	Confirmed

COA #:	SBA-26-0190	Method:	Full QC Panel
Lot Number:	IWR-090926RT1	Analysis Date:	2026-09-13
Labeled Content:	20 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Black #010101

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.96%
Content (per vial)	Report only	21.32 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0190**
Access Code:	**AKELQNSC**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0168

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 30 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-082526RT30	99.93%	31.82 mg	Confirmed

COA #:	SBA-26-0168	Method:	Full QC Panel
Lot Number:	IWR-082526RT30	Analysis Date:	2026-08-28
Labeled Content:	30 mg	Date Received:	2026-08-25
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Royal blue #3223A1

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.93%
Content (per vial)	Report only	31.82 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-08-28

COA #:	**SBA-26-0168**
Access Code:	**FUXDKV4K**
Analyst:	**K. Reyes**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-28

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0169

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-082526RT20	99.97%	21.22 mg	Confirmed

COA #:	SBA-26-0169	Method:	Full QC Panel
Lot Number:	IWR-082526RT20	Analysis Date:	2026-08-28
Labeled Content:	20 mg	Date Received:	2026-08-25
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Royal blue #3223A1

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.97%
Content (per vial)	Report only	21.22 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-08-28

COA #:	**SBA-26-0169**
Access Code:	**2RCY84LU**
Analyst:	**K. Reyes**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-28

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: DPS-5615336
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Report Number:	1020358
Appearance:	White Lyophilized Powder

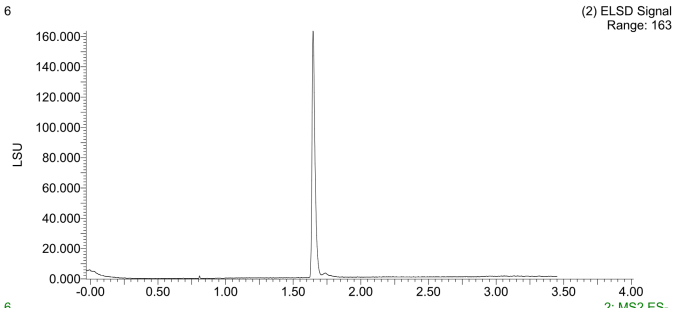
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

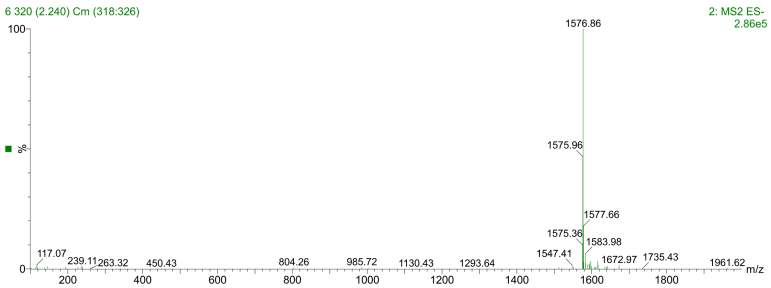
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	10mg	9.93mg	
Purity:	>98%	99.39%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

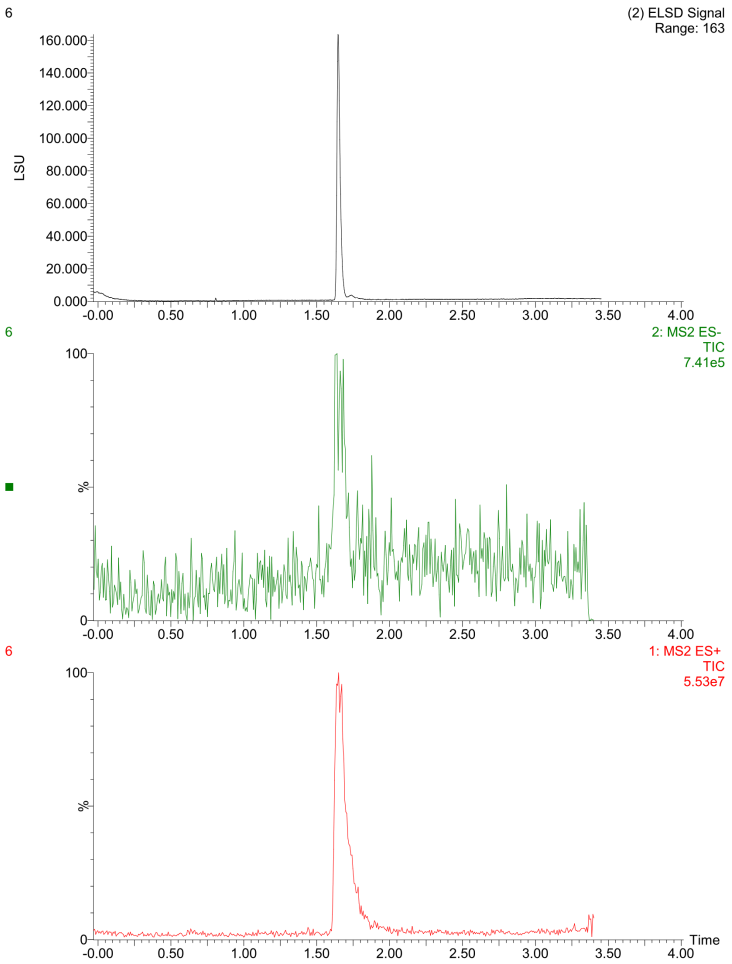
Horizon Analytical

Certificate of Analysis

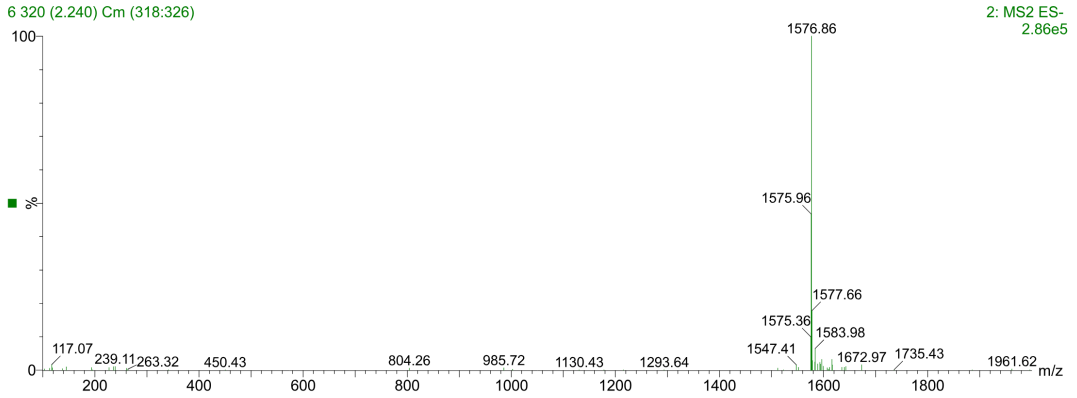
Lot Number: DPS-5615336
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Retatrutide (10mg) • Pubchem CID: 171390338
Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: DPS-5615336
Brand / Distributed by: Iron Within Research
Identity: www.ironwithinresearch.com

Analysis Date: 07/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Report Number:	1020361
Appearance:	-

CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Pass	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0024

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT20-A	99.64%	22.23 mg	Confirmed

COA #:	SBA-26-0024	Method:	Full QC Panel
Lot Number:	IWR-2607-RT20-A	Analysis Date:	2026-08-02
Labeled Content:	20 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
Purple
#760A7D

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.64%
Content (gross)	Report only	22.23 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0024**
Access Code:	**7X93ZFKR**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0049

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR

Iron Within Research

SAMPLE

Retatrutide — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT10-A	99.95%	11.23 mg	Confirmed

COA #:	SBA-26-0049	Method:	Full QC Panel
Lot Number:	IWR-2607-RT10-A	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP

Blue

#0017A8

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.95%
Content (gross)	Report only	11.23 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0049**
Access Code:	**CCUPJRUJ**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0050

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 10 mg

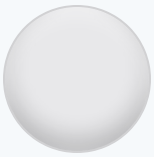
PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT10-B	99.67%	12.06 mg	Confirmed

COA #:	SBA-26-0050	Method:	Full QC Panel
Lot Number:	IWR-2607-RT10-B	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free



VIAL CAP
White
#E8E8EA

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.67%
Content (gross)	Report only	12.06 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0050**
Access Code:	**RJM6R87F**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0051

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TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 15 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT15	99.86%	16.31 mg	Confirmed

COA #:	SBA-26-0051	Method:	Full QC Panel
Lot Number:	IWR-2607-RT15	Analysis Date:	2026-08-02
Labeled Content:	15 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
White
#E8E8EA

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.86%
Content (gross)	Report only	16.31 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.09 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0051**
Access Code:	**KHJ2UWB6**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0052

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT20-B	99.42%	21.12 mg	Confirmed

COA #:	SBA-26-0052	Method:	Full QC Panel
Lot Number:	IWR-2607-RT20-B	Analysis Date:	2026-08-02
Labeled Content:	20 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
Red
#B8232F

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.42%
Content (gross)	Report only	21.12 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.04 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0052**
Access Code:	**ZHWCCFNS**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0053

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
Retatrutide — 30 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-RT30	99.79%	31.76 mg	Confirmed

COA #:	SBA-26-0053	Method:	Full QC Panel
Lot Number:	IWR-2607-RT30	Analysis Date:	2026-08-02
Labeled Content:	30 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
Blue
#0017A8

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	Retatrutide	Confirmed
Purity (HPLC)	>= 98.0%	99.79%
Content (gross)	Report only	31.76 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.08 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0053**
Access Code:	**G6JXA7JY**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: IWR-5859103-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-5859103-P
Appearance:	White Lyophilized Powder

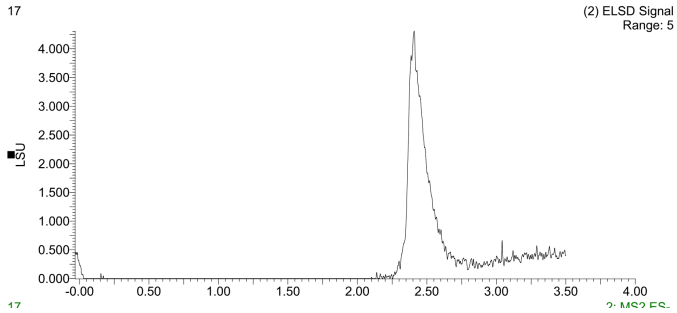
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

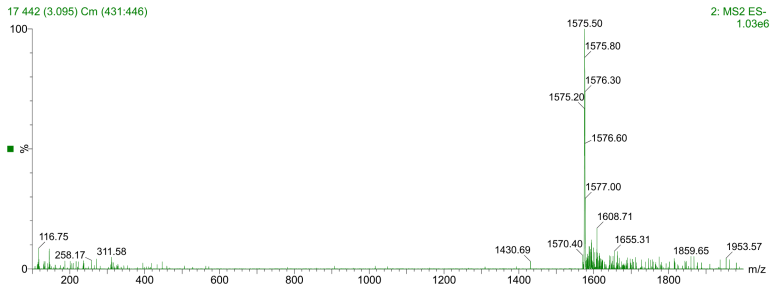
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	10mg	9.84mg	
Purity:	>98%	99.56%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

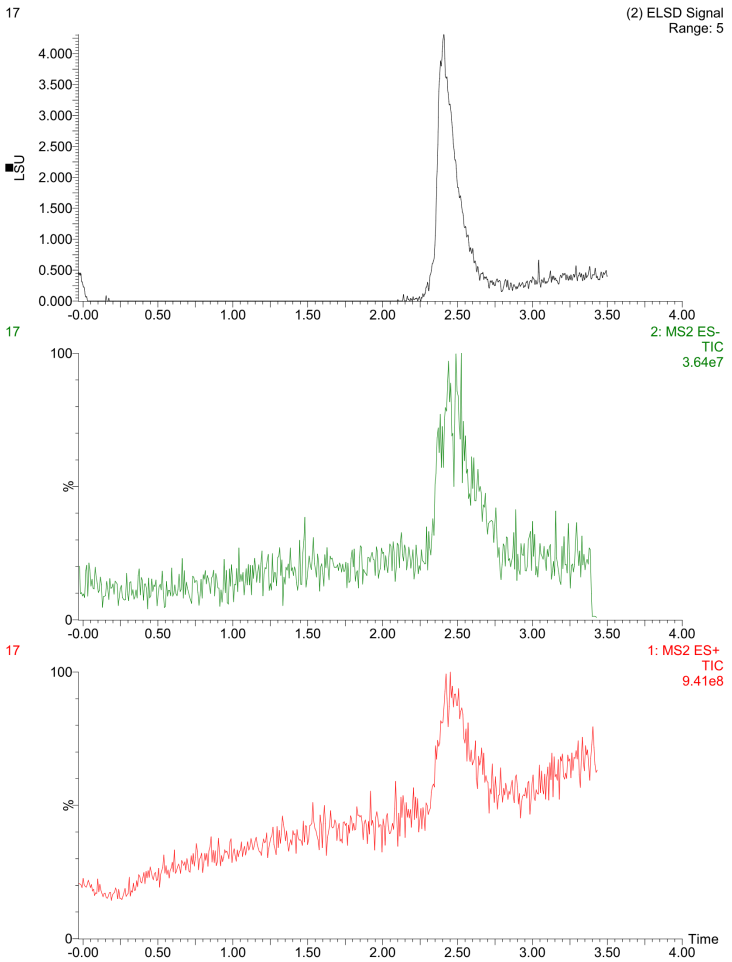
Certificate of Analysis

Lot Number: IWR-5859103-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

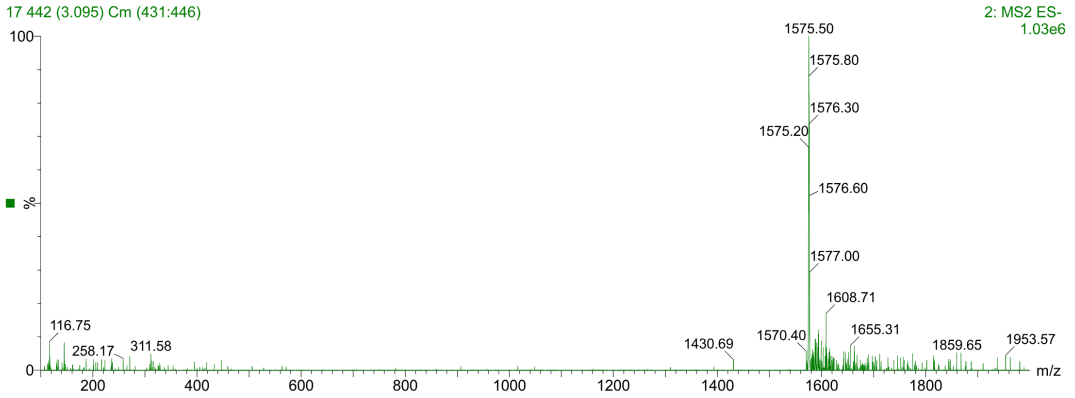
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Retatrutide (10mg) • Pubchem CID: 171390338

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-5859103-E
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-5859103-E
Appearance:	-

CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	-	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

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Horizon Analytical

Certificate of Analysis

Lot Number: IWR-4862753-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-4862753-P
Appearance:	White Lyophilized Powder

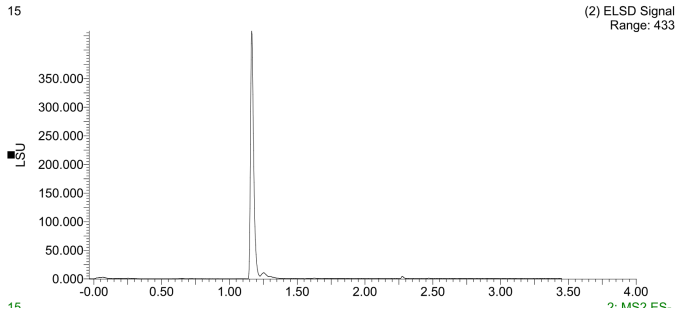
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

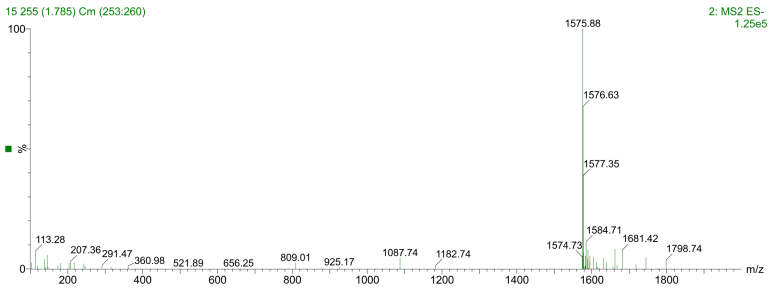
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	30mg	28.9mg	
Purity:	>98%	99.32%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

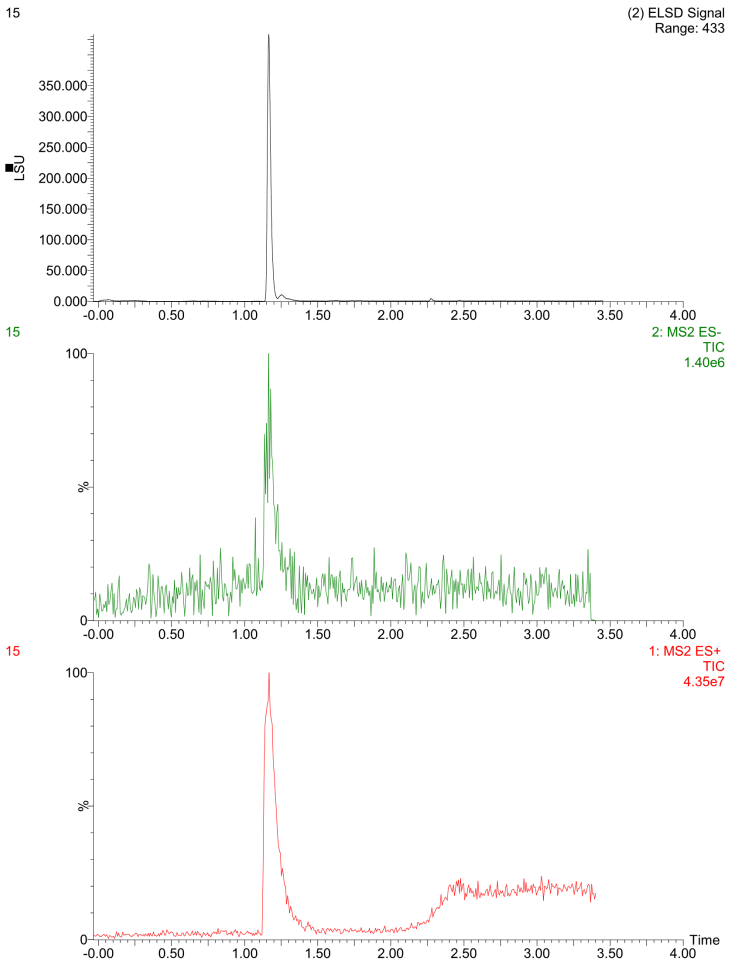
Certificate of Analysis

Lot Number: IWR-4862753-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

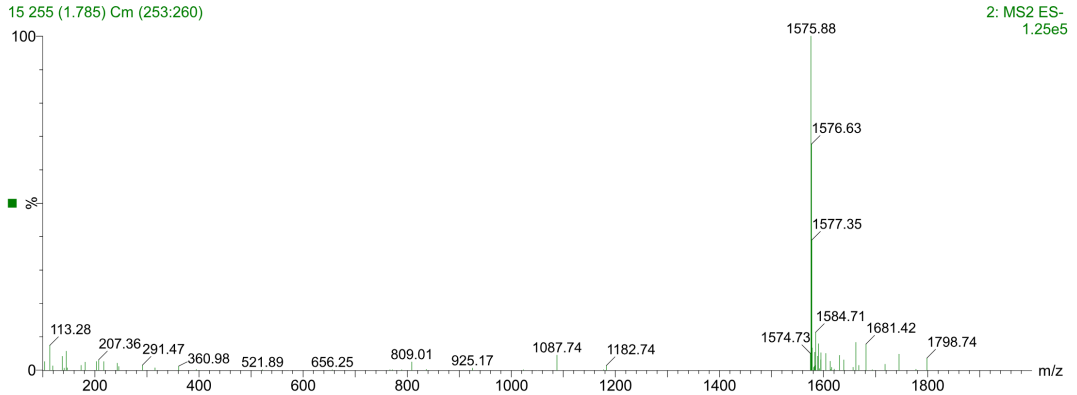
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Retatrutide (30mg) • Pubchem CID: 171390338

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-4862753-E
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-4862753-E
Appearance:	-

CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	-	
Endotoxin:	-	<0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

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Certificate of Analysis

Lot Number: IWR-1764133-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-1764133-P
Appearance:	White Lyophilized Powder

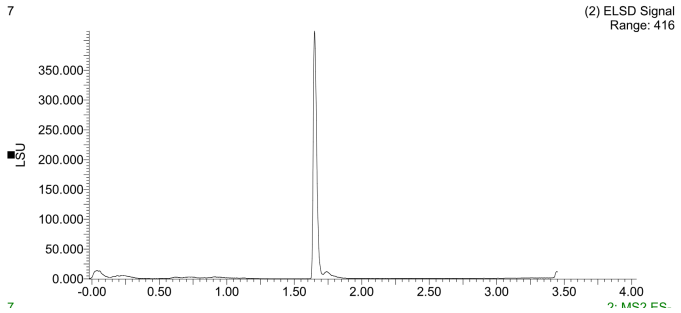
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

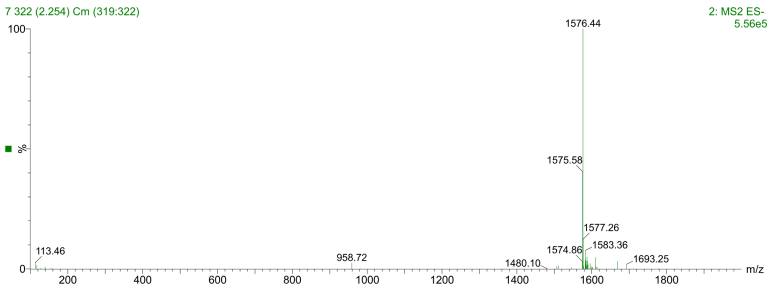
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	15mg	15.05mg	
Purity:	>98%	99.44%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

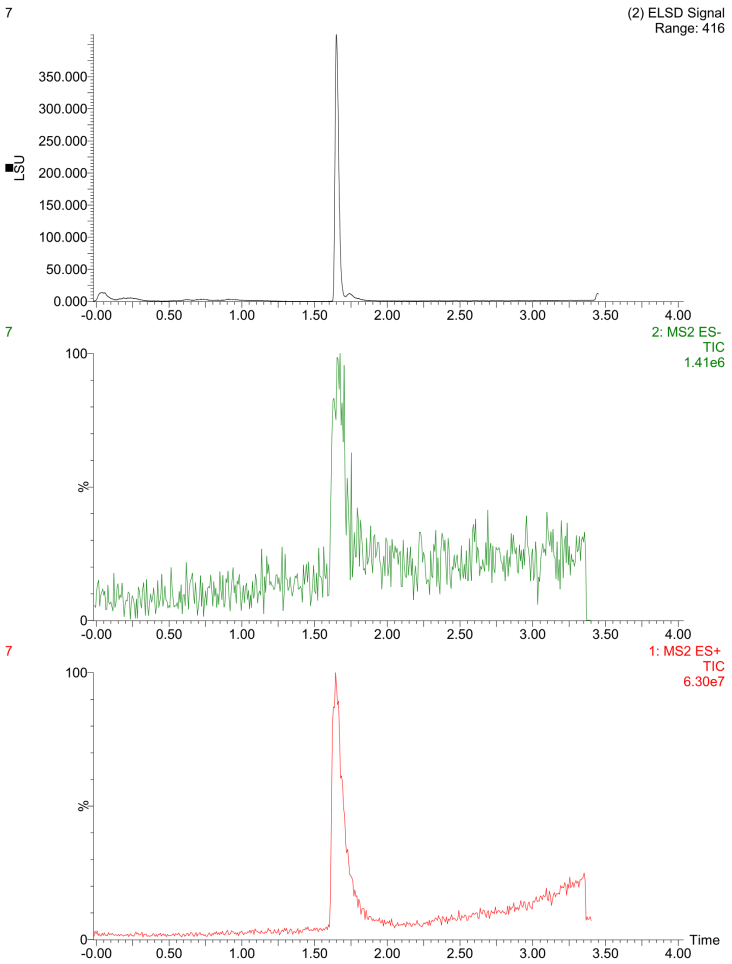
Certificate of Analysis

Lot Number: IWR-1764133-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

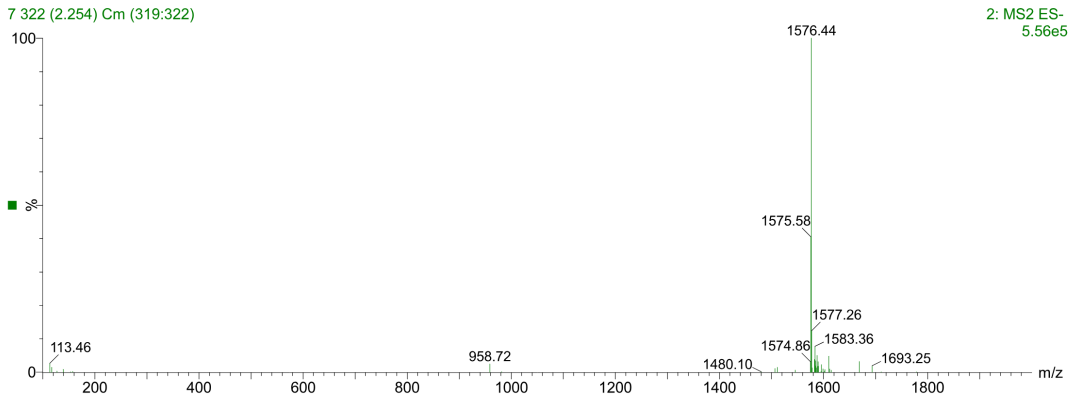
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Retatrutide (15mg) • Pubchem CID: 171390338

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-1764133-E
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-1764133-E
Appearance:	-

CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	-	
Endotoxin:	-	< 0.05 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

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Horizon Analytical

Certificate of Analysis

Lot Number: IWR-5564474-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-5564474-P
Appearance:	White Lyophilized Powder

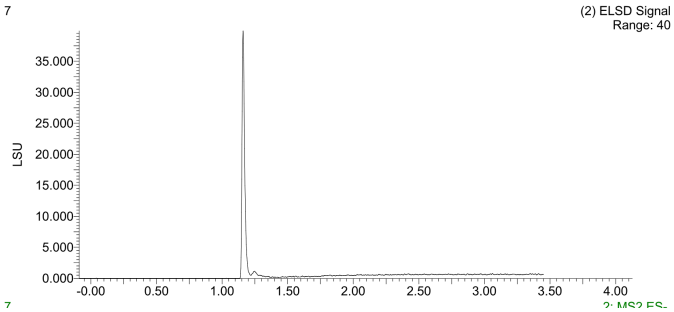
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

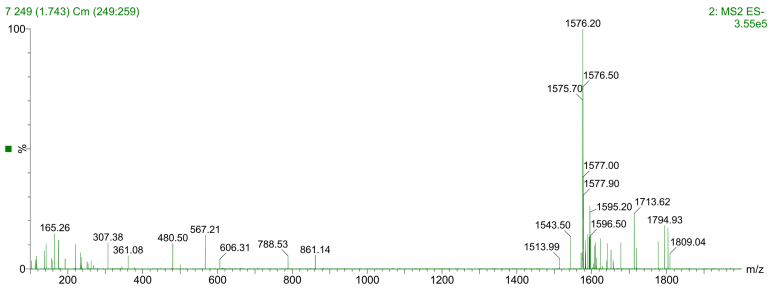
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	15mg	15.11mg	
Purity:	>98%	99.32%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

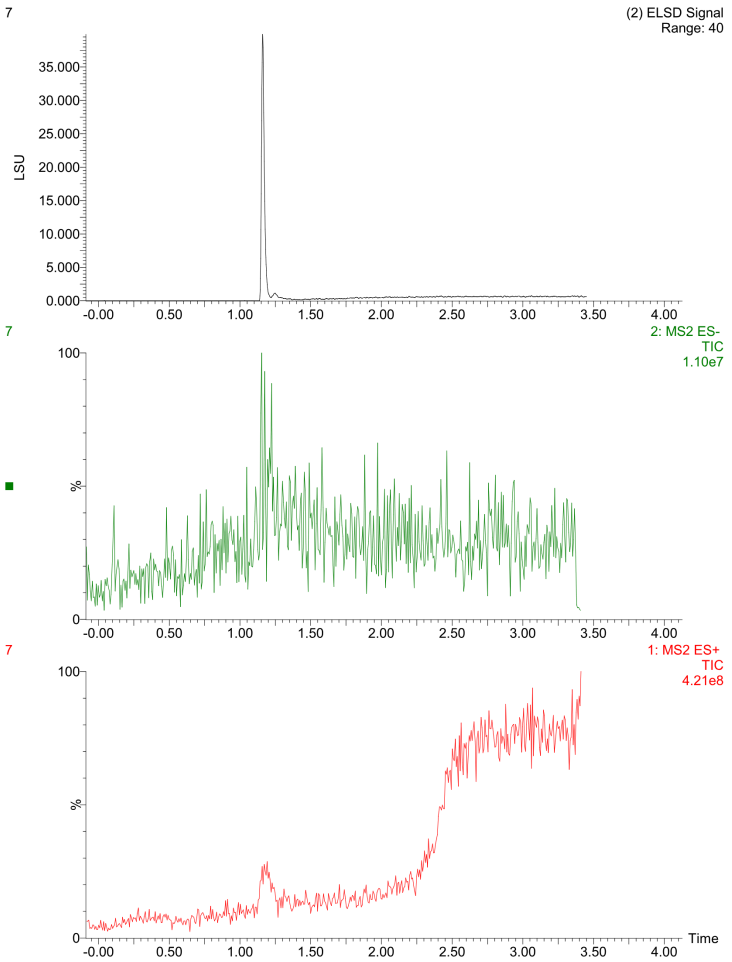
Certificate of Analysis

Lot Number: IWR-5564474-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

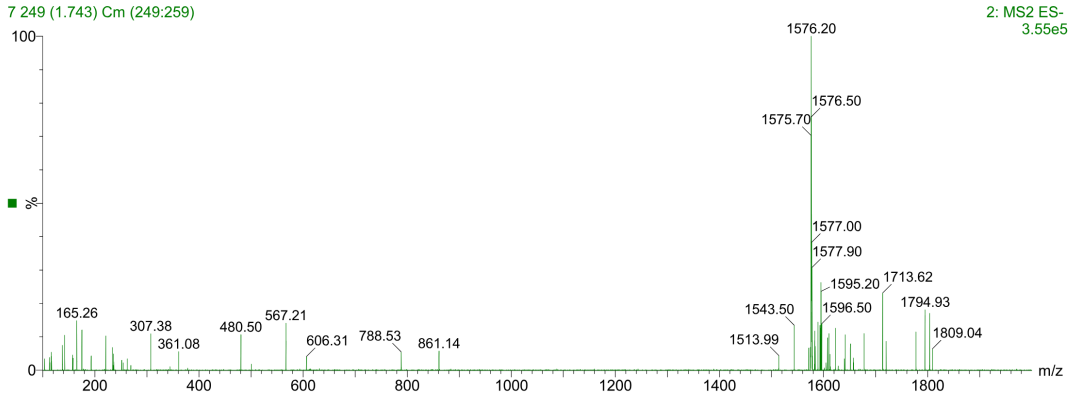
Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Retatrutide (15mg) • Pubchem CID: 171390338

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-8588568-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-8588568-P
Appearance:	White Lyophilized Powder

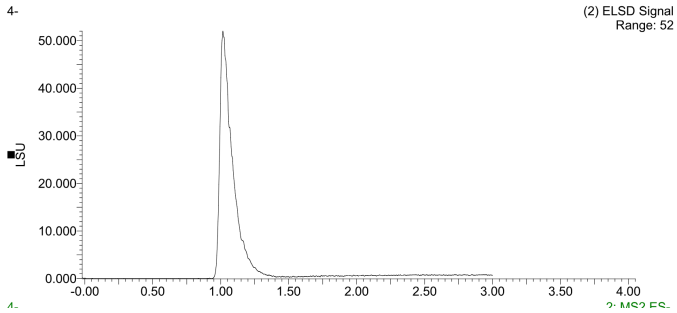
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

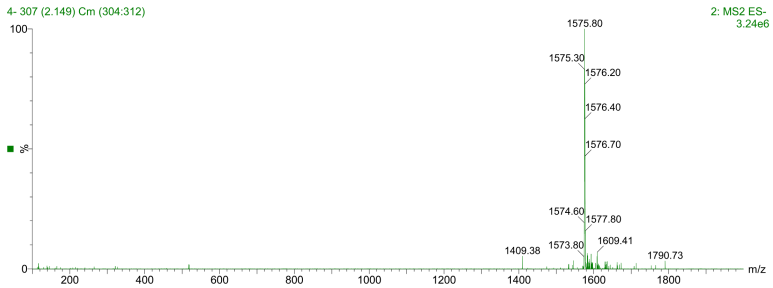
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	30mg	29.5mg	
Purity:	>98%	99.4%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

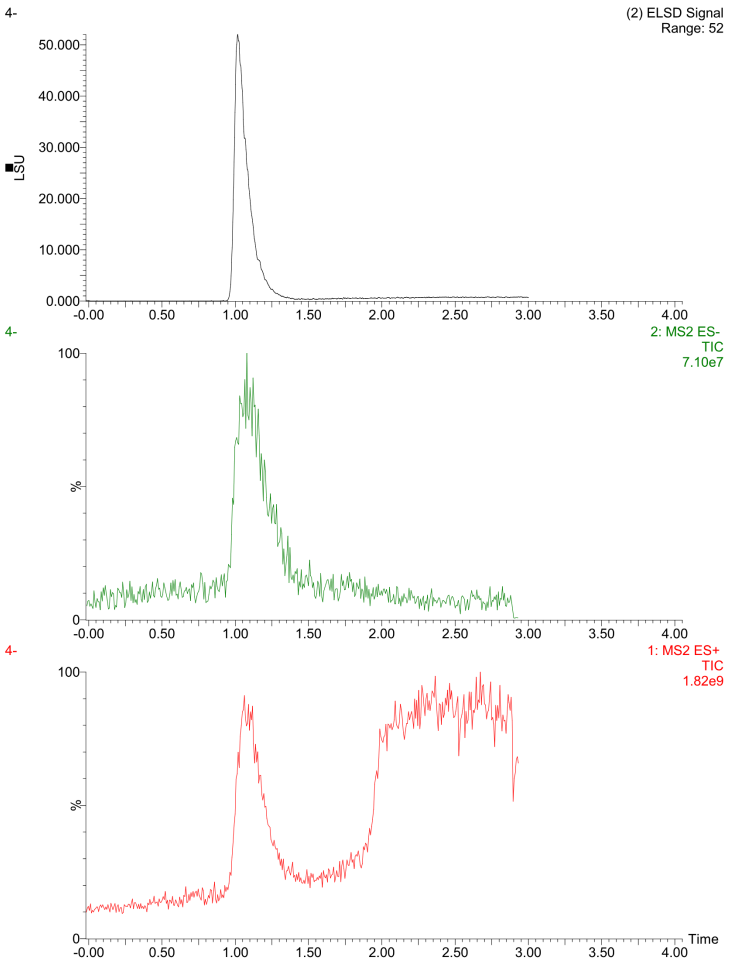
Certificate of Analysis

Lot Number: IWR-8588568-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

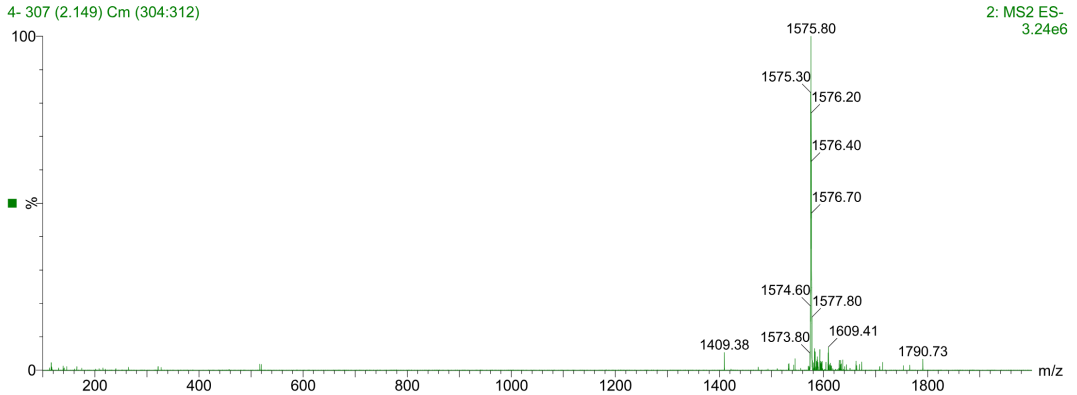
Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Retatrutide (30mg) • Pubchem CID: 171390338

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-9759917-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Compound:	Retatrutide
Lot:	IWR-9759917-P
Appearance:	White Lyophilized Powder

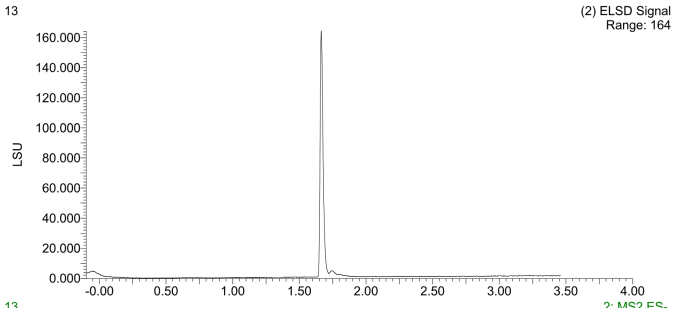
CAS:	2381089-83-2
Formula:	$C_{221}H_{342}N_{46}O_{68}$
Mol Weight:	~4731 g/mol

Pubchem CID: 171390338

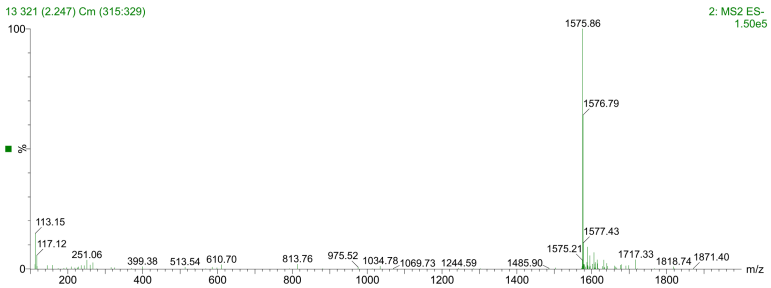
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	Retatrutide	Retatrutide	
Quantity:	10mg	9.82mg	
Purity:	>98%	99.3%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

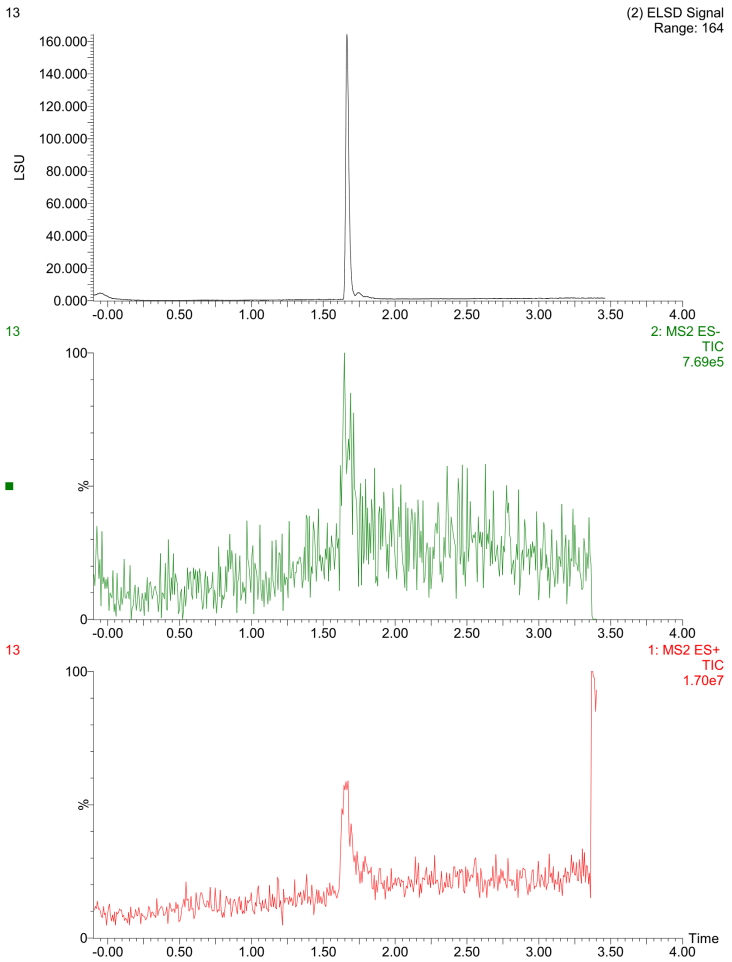
Horizon Analytical

Certificate of Analysis

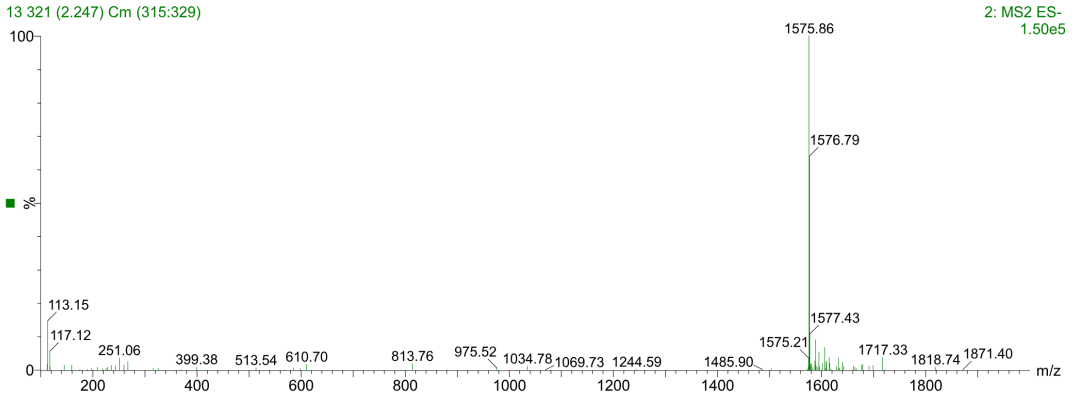
Lot Number: IWR-9759917-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 04/21/2026
Analysis Conducted: 04/16/2026
Searchable via: horizonanalytical.com

Retatrutide (10mg) • Pubchem CID: 171390338
Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Certificate of Analysis

Accession Number	2602110106
Client	Iron Within Nutrition LLC
Search Code	Iron2602110106

Received Date:	02/11/2026
Reported Date:	2/12/2026

FREEDOM DIAGNOSTICS

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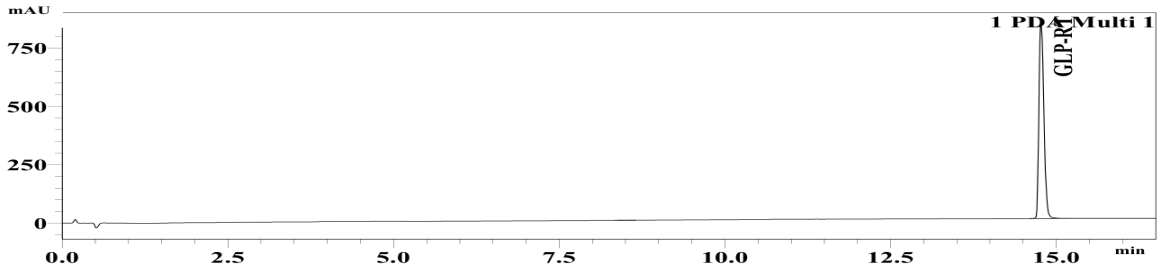
Searchable via: FreedomDiagnosticsTesting.com

Product	Retatrutide 20mg
Net Peptide Content	22.14 mg
Identity	GLP RT

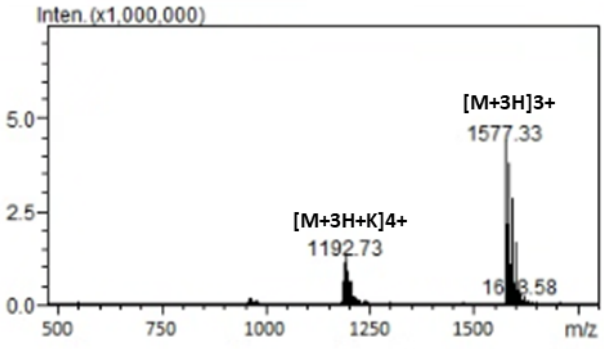
Lot	N/A
Purity	99.946%
Appearance	White Lyophilized Powder

All Chemical Analysis was performed by HPLC with UV Detection Coupled with Mass Spectrometry

Mass Identification	Result
GLP RT	**99.946%**







Stephen Schmidt

Stephen Schmidt

Principal Chemist

COA: 2602110106

The peptide purity analysis reported here was conducted using LCMS/MS under standard laboratory conditions. This analysis is intended for informational purposes only and is specific to the sample(s) provided. The peptides tested are intended for research use only and are not approved for human or veterinary use, diagnostic, therapeutic, or clinical applications. Results should be interpreted by qualified professionals within the scope of the intended research. The accuracy and reliability of the test may be influenced by sample integrity, handling, and other experimental variables.